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Information Technology in support of Living Oriented Society

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abstract: The need of establishing human information technology (HIT), which is aiming to enrich living of the people is proposed. Human information technology should enhance various process of human communications, which are not limited to message transmission. HIT should further promote every member of a family to take broader roles of collaboration, support effective energy saving and resource recycling, and enhance interactions among individuals.
keywords: human communication, information age, energy saving, resource recycling, comfort, network karaoke, image performance

1. Computer Information Technology

Information technology so far might be better named as computer information technology. It started from processing the industrial and business data, and has contributed in improvement of efficiency, in cost reduction, etc. The computer information technology has been the key technology to advance the industrial societies. In the early days of computer information technology, the key problems were the way to reform the business so that they could utilize the computer facility of data processing. Most typically in the countries like Japan, some companies stopped using kanji (Chinese characters) to specify their products and name of workers and began using katakana which was only operable Japanese letters on computers in those days. Computer seemed so effective and useful even worth to abandon their own literacy. This was the ways advanced business men preferred to reform their company.

As the computer improved its capacity, it became capable of processing Kanji. The direction of computer information technology was then reversed. They began adapting the technology to the information systems consistent to the conventional culture. Information technology essentially should be to enhance communication and understanding among people, by using computers and other technical systems.

Same reading (katakana) but different letter (kanji)

カノウ (katakana)
加納, 叶, 嘉納, 狩野 (kanji)

ヒロシ (katakana)
博, 弘, 洋, 浩, 宏, 寛, 裕, 浩志, 弘司, 博司 (kanji)

Many Japanese names sound equal but written differently. When a name is written by the reading (katakana), postmen were confused in delivery, and nurse could not realize who the patiant is.
More over sentenses in katakana was difficult to read.

Fig. 1 Katakana reading and kanji names

According to the classical concept, advancement of the industry is achieved by mass production and automation. They tried to introduce new or high technologies in order to advance the degree of automation and manpower saving. The office automation was introduced to reduce man work engaged in processing routine business data. Encouraged by the preliminary success of data processing, business men began to realize a new possibility of using data processing. Business automation did not reduce the data processing work, but continuously created the work of data processing. Computer market was continuously expanded, not because they succeeded in manpower saving by office automation, but they succeeded in creating new markets and new type of jobs.

People have learned that the mass production is less flexible to the change of market needs and that in order to provide innovative products to the market, human commitment is essential in cooperation with automation. The key for the modern industry is to organizing the human communication with sufficient knowledge to the recent technology among designer, planner, worker, service men and the users. Introduction of scientific measuring systems in factories has enable the workers objectively estimate the cause and effect of various phenomena, and let them understand the ways to prevent errors or defects, and let them improve their process or products.

Introduction of high technology and new machine as well as improvement of the product require creative ideas and problems finding among various sort of people related. Automation does not remove human work, but it changes the quality of work. People have to invent new jobs within or in relation to the new and old work environment.

High technology and new technology have to be applied not to remove human work, but to create new work chance. By making the agricultural machines smaller and easy to operate, old people with no technical back ground could continue working in the field, which is of benefit both to the farmer and for the society.

The POS systems are one of the most heavy duty information systems. The way of using the systems are different by the manager. Most typically the system is used for automatic shop keeping, that is to order the commodity equivalent to the amount of sales automatically. The other way of using the POS systems is to use it to decide the amount of order. So the POS is used as a decision support tool, and the shop keeper might decide what should be supplied. A technical system can be used either for automatic shop keeping or human centered. In the human centered shops, the manager has to analyze the reason for the sales change, and related changes in shop environment. The manager will learn from the daily success and failure.

In the automatic shop keeping, the programmer will be satisfied if the cost for the shop manager could be saved by the intelligent programming. Automatic shop keeping is possible or effective when environments around shops are homogeneous and no competitive shops are located around there.

2. Information Age

Links among a family are changing in the information society. Formerly children learned how to cope with nature in a family. They learned how to cut wood to make fire, or how to protect housing against rain and snow. Now they learn how to cope with economy. They learn the value and the functions of money, and how it changes by the economy. They learn to establish an economy unit in the society. In the near future they have to learn how to cope with the flood of data. The role of the family was to provide physical and economical shelters for their children. In the information age it is to provide shelters against the flood of data and to bring up the problem finding and problem communicating competence to their children. In the former times it was valuable to have a copy of data, but now it is rather easy to make a copy. It is nonsense for an individual or for a business organization to keep large storage of data. Of the essential significance is to select necessary data and throw away less significant data. Selection of necessary data out of flood of less significant can be done only by having proper understanding to the problems. Roles of HIT are to support the family to bring up and to enhance the problem finding capability of the young, the adults as well as the aged.

Education in advanced industrial countries are still based on the memory training. They are trained how to read and write in the primary school, trained to remember the conjugate of irregular verbs, trained how to solve second order equations in the high school, and again in the computer course at university, learn correct use of command languages. They are continuously trained to solve problems, as quickly as possible without making the mistake. But now a days computer can solve many problems, if properly defined. And robots can make camera, or radio and many more thing better than the man. It is of worthless for the man to compete with computers

or robots, in the tasks which machines can do. The task of the human worker in the highly robotized factory has been to find a new problems in the factory, in the product and in scheduling. The problems have to be discussed within the shop floor and between different work sections. The workers have to be trained to communicate with various levels.

Communication in limited sense was the transmission of message. Various technical tools have been introduced to support clear and exact message transmission, such as microphone and speaker, slide projector, OHP and copy machine. Communication, however is not a mere transfer of message. Communication is the process of mutual understanding and reaching consensus. To promote mutual understanding, people has to be aware of the barriers of communication. There might be various barriers of communication, distance, time, language, knowledge bariiers. Communication may comprise such processes as finding the opponent of talk, specifying the problems, catching the chance to talk, preparing proper expression, follow up commitments, etc. [Tamura, 1990]. In evaluation of media communication not only the message transfer, but also the whole processes of communication should be examined whether they are properly supported.

Narrow sense
Communication = message transmission
Broader sense
communication = mutual understanding.
communication technology =
to find out the barriers of communication
and to remove them

Processes of communication
search for whom to contact
refine problems
catch chance
proper expression
transfer or redirect
follow up

Fig.2 Concept of communication

3. Everything, Everywhere

Computer work station has been proud of being occupating the major area of desk top in the office and other work places. In the near future, the size of computer will be as small as pocketable or wrest mountable and associated display will be mountable on head like eye glass. With such predictions in mind, we have to think of the information technology tightly involved in the daily life of the people.

Small computer is not a big data center nor all mighty in handling multimedia. They are transferring media data to the center near by. They should be small but reliable agent to access centralized or distributed data base. As mentioned previously, communication is not mere transfer of message. It start from finding the person to communicate. Since people are moving more frequently and larger distance, and engaged in diversified tasks, the service to find proper person and provide appropriate media of communication become significant.

Small computer is beautiful because it could be installed in every piece of articles in the kitchen, living room, bath room, etc. It will be installed in walls, celling and floor as well as movile phone, and AV machines. They should be linked together by the wireless, and could be accessed every body at home and out side. In order to assure free and flexible access, member of family including pets and plants, as well as various sensors and transmitters installed in machine and walls shall have addresses.

Machine like kitchen oven, washing machine, VCR are normally monitored by specific person today at home. When a programmed task is complete, they make some message sound to acknowledge the completion to the people near by. It is not necessary to make message sound to be heard by everybody. In house locator will sense the location of the particular member of family and transfer the message sound to the person. If the locator detects the person is out or not in accessible locations, the message should be transferred to alternative person near by. Of course if necessary, the system will try to reach other responsible agents.

The address systems today are not convenient, when people begin to use several access number. Some people already have several addresses, which are fixed to location (home, office, car,etc) or limited function (phone, facsimile, telex, video phone). It is inconvenient, because people have to remember many addresses to access same person. It is more convenient if a person could be accessed by a same address. The sender will forward person access code and certain function code that will specify his selection of which communication media(phone, facsimile, telex, video phone).

The recipient terminal will analyze the function code and transfer the call to the terminal available to the recipient, depending on the location and situations. The access control communication and media

communication will be clearly differentiated. Even in case of normal telephone, it takes considerable time to access the person to whom to call. Normally the phone call comes most frequently to the person who are out of office or too busy to answer the call. When a recipient is busy and unable to answer the call, the call might be transferred to fax or voice recorded terminals. If the recipient is attending to a conference, and not be able to answer a phone call, access control agent will leave message where the fax or voice recordings are stored.

Sensors, transmitters and movile networking will be especially useful to help people with special needs. They are often not capable of sending commands punctually. For such people network agent will analyze the behaviors of the person, and try to understand his intention. The system will try to manage the situation automatically based on the understanding.

4. Comfort of Coordination

Now a day, a society might be regarded poor, not because of absolute lack of goods but by the lack of flexible delivery systems. People like to live on their own preference, and they have to find ways of coexisting with the people of conflicting demands and preferences. At home, people of three generations are often living together. The aged and the adult, or the children like and need to have different foods. Or the meal time of is different by person in a family and different meal is served upon each demands or preference. Recently it is not by no means strange that each individual on the dinner table is eating different food. The preference changes irregularly depending on the previous menu and physical conditions. It is not bad to have different food in different time independently, but it is also enjoyable to have food jointly together. People like to have such diversity of living independent and interacting together.

To keep such mutually independent but coordinated living, no individual should be constrained in the house or kitchen keeping. The house hold or cooking machines have to be conveniently operatable to many of the family members including aged and children. It is not appropriate to design these machines assuming the principal users to be the young ladies. It is also not appropriate to to think the main objectives of machine use is to minimize the house hold labor. The machines have to be designed to increase the freedom of use and non-use, freedom to adapt various members of the family to create new possibilities. It is most desirable to create a new chance of communication and collaboration by introduction of a new machine. Word processors created chance to write sentences for many people, why not kitchen processor create a new food communication. Kitchen processor is not the tool to automate food processing, and to minimize the cost. In stead it is the tool to reform the family life colorful and rich. It should be pleasant enough like game computers or karaoke machines. Because it is fashionable and pleasant school children like to cook, husband proud to cook for their family.

5. Resource Recycling

Wealth of a nation used to be measured by the amount of production, or by the number of owners of such goods like car, telephone and TV, or by the amount of energy and resource consumption. It is not the amount of goods, but the coordinated use of the goods which is essential to define wealth of the nation in the information age.

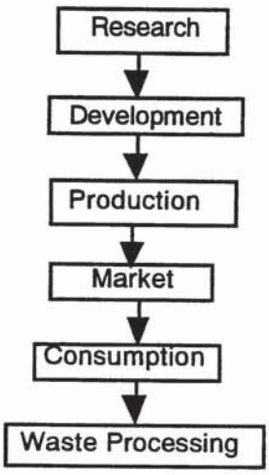


Fig.3 Resource flow in the industrial society

Human information technology (HIT) shall provide effective tool for building ecologically sustainable societies. Information technology should be used more effectively to prevent over consumption of energy and resources. In the industrial ages a lot of effort have been done to make manufacturing process effective and less resource consuming. But the delivery as well as consumption process was by no means scientific. Information technology has to be based on the living, in stead of production. So far research and development were more interested in innovation of production process. For the effective and comfortable use of resource and energy, studies on the living of the people are most essential. Fig.3 shows the flow new technology in the industrial

societies. The scientific findings and technical inventions gave influences to the living in one direction, from up to down. The science and technology in the living oriented societies have to be situated near to the living, and have to consider the recycling process as well as the production, as shown in Fig. 4.

In collection and classification of used goods, higher information technology is required than in the production process. The sales market of the used goods needs more information technology than the sales market of the new product, since the used goods are different one to another in every detail, and often require some repairing. The users of used goods might have much diversified requirements, and the users have to check the match to his requirements. The collected used goods have to be classified into reusable and not reusable. The used goods have to be transferred to the delivery market. From the not reusable goods , reusable components should be decomposed to be used as parts. The decomposed goods shall again be separated by different materials, like paper, plastic, iron, copper, aluminum, gases, etc.

For the collection, decomposition and separation robots and information technology should be introduced, in the same way in the production.

Efficiency of the resource recycling is essentially determined by the purity of the recycled materials. Toimprove the quality of resources, citizen's consciousness and their participation is most important. This is something related to the worker's participation in manufacturing.

Actual recycling process is by no means effective nor elegant, so the some people do not like to do, if permitted.

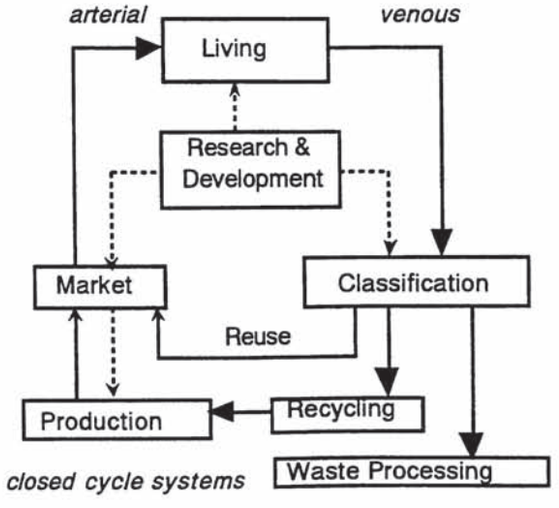


Fig. 4 Resource flow in the living oriented society

Works not beloved by people should be automated. For collecting used paper for recycling, the fax paper should be careful separated in order to preserve quality. At present, people separate it visually. Such use of manpower is by no means effective. It is most desirable to invent a physical sensor to detect fax paper from normal. Even if the sensors are not invented, necessary robot has to be developed. One possible option is to print invisible bar code on each sheet of paper, and the robot will identify the code and separate it. For such purpose automatic machine will most properly used. Developing a recycle robot is tough challenge, since the processing cost should be much lower than in the POS systems or in game computers. But information technology have to challenge continually to the field where social demands exist.

HIT will contribute in developing the clean and automatic recycling process, so that people feel like to participate in the recycling activities spontaneously on the one hand, and in making it enjoyable by reforming the activities more fashionable, or by including some game features and fun.

6. Enjoyable Interaction

In the early days of computer information technology people had to adapt to the technology. Thanks to the advances in computers and digital networks, human information technology tightly connected to the living of people are now feasible to grow. Human information technology is not a mere tool of manufacturing, nor business management. it is now an essential part of our living environment. Information technology to be implanted in the social environment should be lovable, enjoyable and well understandable to everybody, while it is safe, reliable and resource saving.

Japanese gardens are beautiful, they look natural. But it is an artificial nature built in a small part of nature. The artificial nature is beyond nature. They provide ever lasting relax and relief. Actually they are always changing by season and by growth of plants and grass, and by rain and snow. Also mind of the people changes. Nevertheless the nature beyond nature are beloved beyond the changes. People now expect for the human information technology such interfaces like Japanese garden.

As the computer information technology advanced, people are required to remember more and more commands, to monitor flood of the data carefully, to operate key or handles in regular order, and to watch

carefully to detect errors. In spite of advances in computer information technology, people have to read more and more papers. In spite of advanced network technology, people have to move here and there more quickly. Now people are no longer interested in high speed and high precision computations. People are more interested in more relaxed and peaceful living environment. They are attracted by the computer use with wonder and surprise. People who are not currently the user of computers, will become the use only when the information systems could provide such environment, where people share the experience of tears and laugh with and through the environment.

Among various computer technology screen savers are the one providing rest and relief. Some of them make use of full computer power to colorful artistic motion. In place of gift articles exchanged by friends and family, computer gifts will be exchanged among people in future. Computer gifts are like a flower in the room, they will decorate the computer screen. When computer works are not busy they come out and present beautiful screen performance and melody. If the user interact with the gift program, it learns the preference of the users and adapt its performance to the preference.

Each country and people love each culture. One country may prefer top hero, and top sales. Other country may enjoy a culture they could participate together. In Japan, there have been a music movement which encourage every one to sing. It started from NodoJiman, MittsunoUta, Utagoe and to Karaoke. Popular singers will present the time catching ways of singing to a song. The player who are by no means trained in singing, challenges to sing the song he loves. The audience send the applause to his/her participation and contribution to make homely environment. A lot of electronics have been introduced to enhance Karaoke performance. They normally use laser disc for prompting words of song on the display together with the back music. In the most recent Karaoke, called network Karaoke, they use ISDN digital service to send MIDI code and words from computer located hundreds of kilometers far. At present ISDN service are available in most part of Japan, but the actual user is still not much. Under such circumstance, the network Karaoke is the top user of the ISDN service, which was never expected by the network investors.

Players like network Karaoke, because the new titles of songs are included in the list quicker than the disc Karaoke. The time for publishing the optical disc is too long to compete with network Karaoke. In case of network Karaoke, as soon as the code of a song is registered in the central computer, the song can be played everybody everywhere. At the same time Karaoke studio need not have to stock big file of optical discs.

Finally future of the Karaoke has to be predicted. Player want to pretend himself virtual environment. Player might want to sing it at the Waikiki shore, or in Pat O'Brien house in New Orleans, or in the front of Heian Shrine in Kyoto. Multi media technology will enable him sing in front of the back ground image reproduced from optical video.

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